PROJECT DOCUMENTATION

An AI-Based Personal Finance Advisor Bot

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Department of Computer Engineering
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Submitted By

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1. Project Overview

1.1 Introduction

The Personal Finance Advisor Bot is an AI-powered financial planning assistant that helps individuals take control of their personal finances with clarity and confidence. It provides a single digital environment in which a user records monthly income, logs daily and weekly expenses by category, receives personalized budget plans, and tracks saving progress. The web application carries the interface name **FinAdvisor AI**.

The platform combines a Flask and SQLAlchemy backend with an AI engine .

1.2 Scope of the System

The system is a scalable full-stack web application intended for individuals who manage personal budgets. The project brief describes four representative user groups: salaried professionals, college students, freelancers with variable income, and household managers. By automating budget generation and giving structured saving recommendations, the system reduces the effort of manual financial tracking.

1.3 Project at a Glance

Attribute	Detail
Project name	Personal Finance Advisor Bot
Interface name	FinAdvisor AI
Application domain	Personal finance management and budget planning
Target users	Salaried professionals, college students, freelancers, household managers
Interface modules	Dashboard, Income, Expenses, Budgets, Goals, Reports, AI Advisor
Currency shown in interface	Indian Rupee (INR, ₹)
Backend / data	Python, Flask, SQLAlchemy, SQLite
AI engine	GeminiAI
Frontend	JavaScript
Project record	8 epics, 15 tasks, 0 subtasks

2. Problem Statement

2.1 Problem Description

Individuals who manage their own budgets must record income, categorize every expense, compare spending against income, and decide how much to save. When this is done manually it is time-consuming, and overspending often becomes visible only after the month has ended. The Personal Finance Advisor Bot addresses this by recording the data in one place, analysing it, and returning budget plans, saving suggestions and a structured monthly report.

2.2 Challenges Addressed

The project brief identifies the following challenges, each linked to a user group:

- **Locating overspending.** A salaried professional needs to know which spending categories exceed what is affordable, and how to reduce them.
- **Constrained budgets.** A college student must divide a limited monthly allowance between essential and discretionary spending.
- **Irregular income.** A freelancer earns variable amounts from multiple clients and needs recommendations that adapt to fluctuating income and to months with reduced saving capacity.
- **Shared finances.** A household manager must monitor family income and expenses across groceries, utilities, education and healthcare, and detect category-level overruns.

3. Objectives

3.1 Primary Objectives

1. To allow users to record monthly income and to log daily and weekly expenses by category.
2. To generate personalized budget plans from the recorded financial data.
3. To provide AI-driven spending insights and structured saving recommendations.
4. To track saving progress within a centralized digital environment.
5. To produce a structured monthly financial report.
6. To build the platform on a scalable full-stack architecture that can accommodate future enhancements.
7.

3.2 Objective-to-Capability Mapping

Objective	Supporting capability in the system
Financial data recording	Income entry with source and description; expense logging by category
Budget generation	Category budget limits, including an Auto-Generate (50/30/20) option
Spending insights	Budget utilization status per category; AI Advisor assessment
Saving management	Net savings and savings rate in the monthly report; saving recommendations
Reporting	Monthly Financial Report with print / export to PDF

4. Proposed Solution

4.1 Solution Description

The proposed solution is a web-based financial planning platform organised into seven interface modules, which appear in the application's navigation bar. Users enter income and expenses; the application computes budget utilization and monthly figures; and an AI Advisor component produces a written assessment. Flask serves the application, SQLAlchemy manages database access to SQLite, and GeminiAI supplies the AI-driven analysis.

4.2 Functional Modules

The table lists each module visible in the application together with what the supplied screenshots show about it. Details that could not be confirmed from the material provided are marked accordingly.

Module	Purpose and evidence	Reference
Dashboard	Overview page with expenses by category, six-month cashflow trend, budget utilization and recent transactions	Figure 10.1
Income	Income history table (date, source, description, amount, edit and delete actions), summary cards and income-by-source panel	Figure 10.2
Expenses	Expense logging by category (project brief); reachable from the Dashboard. Page screenshot	—
Budgets	Category limits, real-time utilization, status labels, Auto-Generate (50/30/20) and Set Category Budget actions	Figure 9.1
Goals	Present in the navigation bar; page behaviour	—
Reports	Monthly Financial Report with financial metrics, AI assessment, expense and income breakdowns, and Print / Export PDF	Figure 9.2

Module	Purpose and evidence	Reference
AI Advisor	Present in the navigation bar; a dedicated page screenshot is An AI Advisor assessment appears in the monthly report	Figure 9.2

4.3 Implemented and Future Scope

The functionality described in the project brief and confirmed by the screenshots is treated as current functionality. The four enhancements listed in the brief are treated as future scope and are detailed in Section 14.

Status	Capability
Current	Income recording; expense logging by category; category budgets with utilization status; Auto-Generate (50/30/20) budget option; monthly report with net savings, savings rate and financial health score; AI Advisor assessment; Print / Export PDF; dashboard visualisations
Future (proposed)	Predictive spending analytics; investment suggestions; goal-based savings tracking; advanced AI-driven financial health monitoring

5. Project Scenarios

The project brief defines four usage scenarios. Each is summarised below.

5.1 Scenario 1: Salaried Professional

A salaried professional logs monthly income and daily expenses across categories such as rent, food, transport and entertainment. The system analyses the spending, identifies overspending areas and generates a personalized budget plan with actionable saving suggestions. At the end of the month the user receives a summary showing income versus expenses, savings achieved, and financial goals for the following month.

5.2 Scenario 2: College Student

A college student manages a limited monthly allowance across essential and discretionary categories. The platform helps set realistic budget limits, track spending habits and receive AI-powered recommendations for reducing unnecessary expenditure, which supports financial discipline within a constrained budget.

5.3 Scenario 3: Freelancer

A freelancer with variable monthly income logs earnings from multiple clients and tracks project-related expenses. The system adapts budget recommendations to fluctuating income, highlights months with reduced saving capacity and suggests strategies for building an emergency fund.

5.4 Scenario 4: Household Manager

A household manager monitors shared family income and expenses across groceries, utilities, education and healthcare. The bot generates a consolidated monthly report, identifies budget overruns in specific categories and provides tailored saving strategies for family financial goals.

5.5 Scenario Comparison

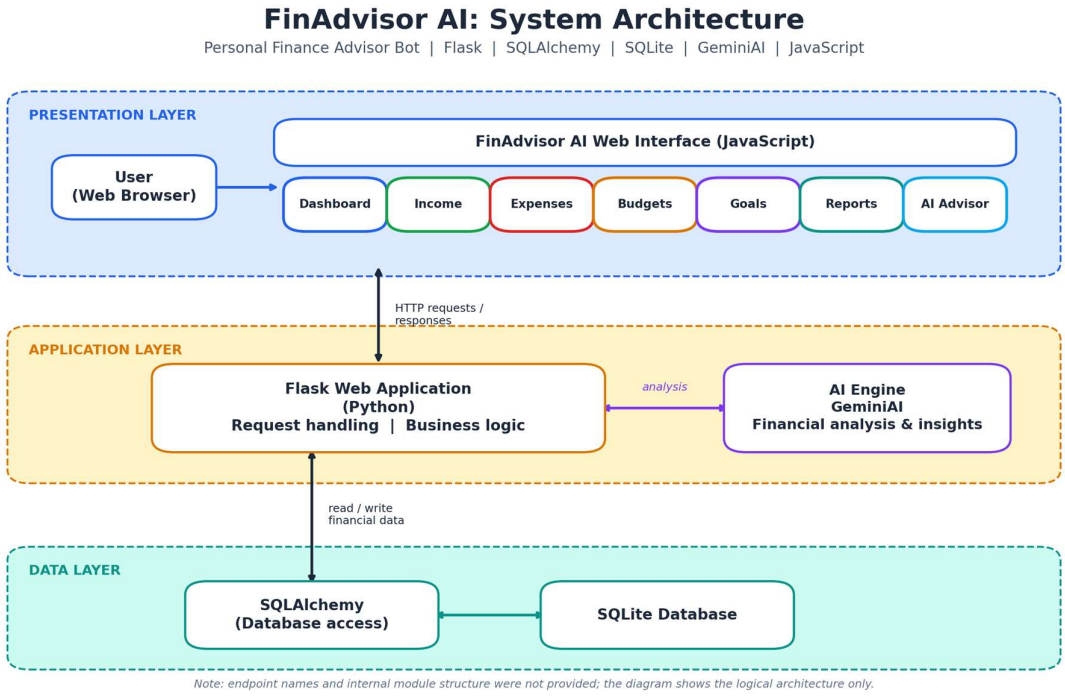
Aspect	Salaried professional	College student	Freelancer	Household manager
Income pattern	Monthly income	Limited allowance	Variable, multiple clients	Shared family income
Key output	Overspending analysis, budget plan	Budget limits, reduction tips	Adaptive advice, low-saving months	Consolidated report, category overruns
Main benefit	Actionable saving suggestions	Financial discipline	Financial stability	Household planning

The interface data captured in the screenshots records income under four source types (Salary, Business, Freelancing and Other) and applies category budgets with overspending status, which corresponds to the multi-source income and overspending-detection themes of Scenarios 1 and 3.

6. System Workflow and Architecture

6.1 Architectural Overview

The platform follows a three-layer full-stack design built from the technologies named in the project brief. The presentation layer is the JavaScript web interface through which the user reaches the seven modules. The application layer is a Flask web application that handles requests and communicates with the AI engine (GeminiAI). The data layer consists of SQLAlchemy, which provides database access, and a SQLite database. Endpoint names and internal module structure are so the figure shows the logical architecture only.



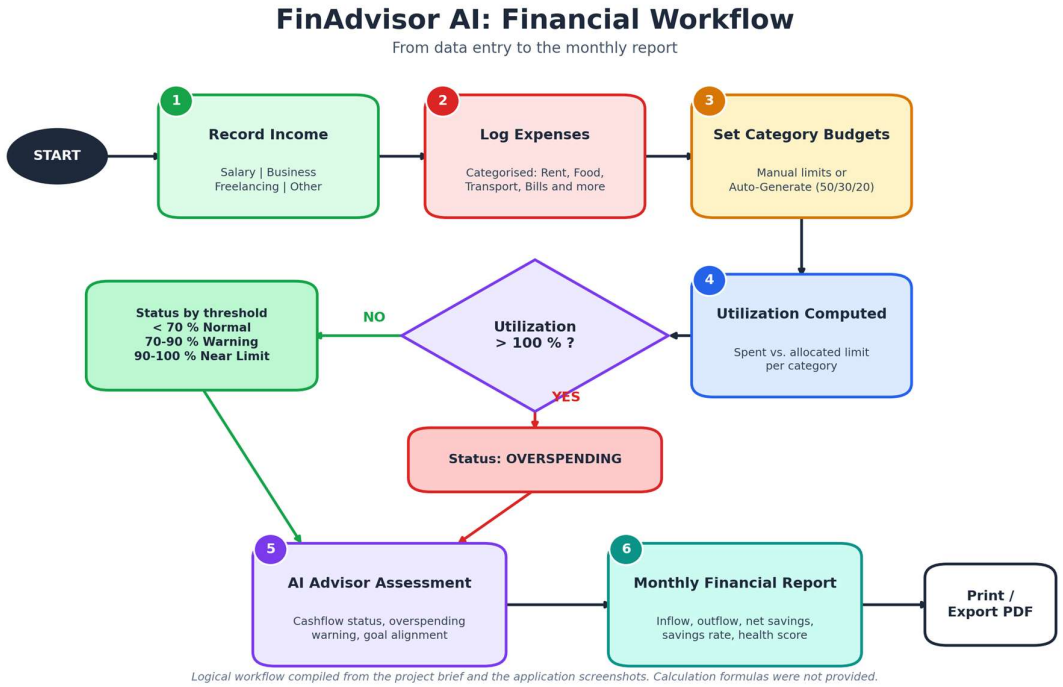
6.2 Component Responsibilities

Component	Responsibility
JavaScript web interface	Presents the Dashboard, Income, Expenses, Budgets, Goals, Reports and AI Advisor modules to the user
Flask application	Web framework that receives user requests and returns responses
SQLAlchemy	Database access layer between the application and SQLite

Component	Responsibility
SQLite	Persistent storage of the financial data
GeminiAI	AI-driven financial analysis and insights

6.3 Workflow

The workflow below follows a user from data entry to the monthly report. It is compiled from the project brief and the application screenshots.



1. The user records income under a source type (Salary, Business, Freelancing or Other).
2. The user logs expenses, each assigned to a category.
3. The user sets category budget limits manually or uses the Auto-Generate (50/30/20) option.
4. The application computes utilization for each category and assigns a status: below 70% Normal, 70–90% Warning, 90–100% Near Limit, above 100% Overspending.
5. The AI Advisor produces a written assessment covering cashflow status, overspending warnings and goal alignment.

7. Technology Stack

7.1 Technologies Used

Technology	Category	Role in the project
Python	Programming language	Backend implementation language
Flask	Web framework	Serves the application and handles requests
SQLAlchemy	Database toolkit	Database access for the application
SQLite	Database	Stores the financial data
GeminiAI	AI engine	AI-driven financial analysis
JavaScript	Programming language	Frontend functionality

7.3 Software Requirements

Component	Requirement
Operating system	Windows 10 / 11, macOS (Monterey or later), or Linux (Ubuntu 20.04+)
Web browser	Latest version of Google Chrome, Mozilla Firefox or Microsoft Edge
IDE / code editor	Visual Studio Code (recommended) or any preferred IDE
Version control	Git (latest version, installed and configured)
Additional tools	Python 3.8 or above; AWS CLI (latest version)

8. Database Design

8.1 Database Technology

The application stores its data in a **SQLite** database and accesses it through **SQLAlchemy**. Table names, column names, data types and relationships were not supplied, so they are not stated here.

8.2 Data Elements Evident from the Interface

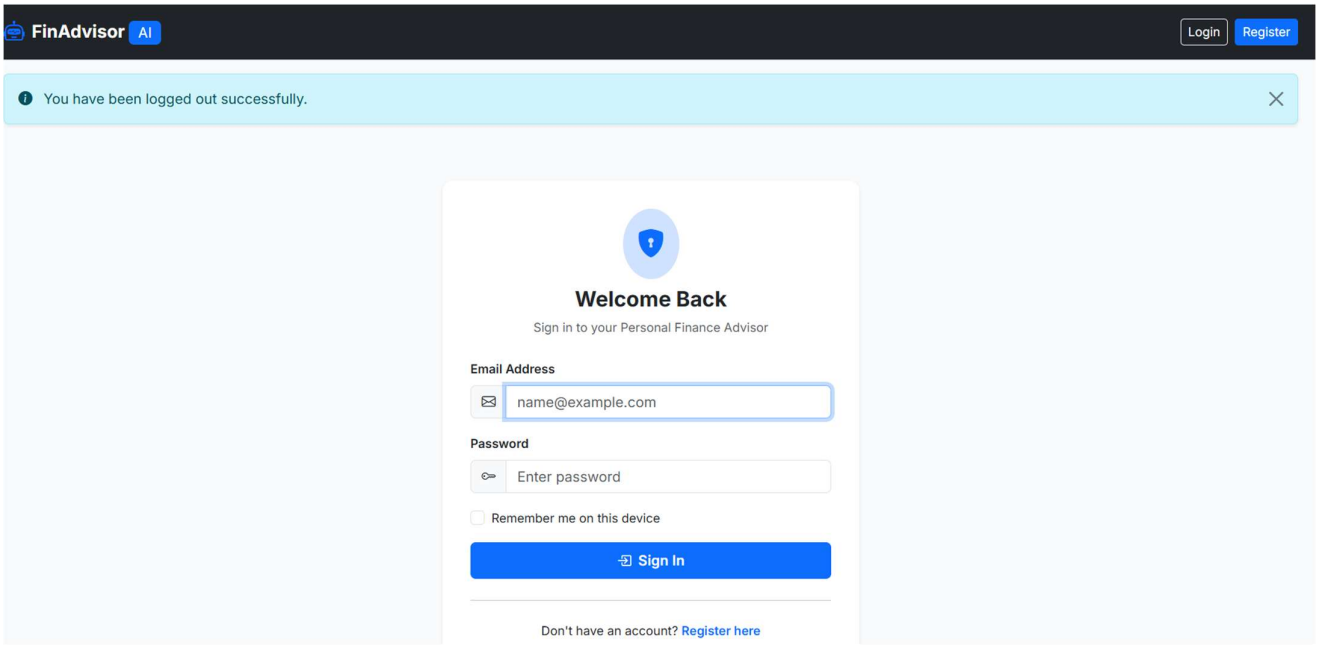
The table lists the categories of data visible in the interface and the attributes the screenshots display for them. These are interface-level observations and may not match the stored column names.

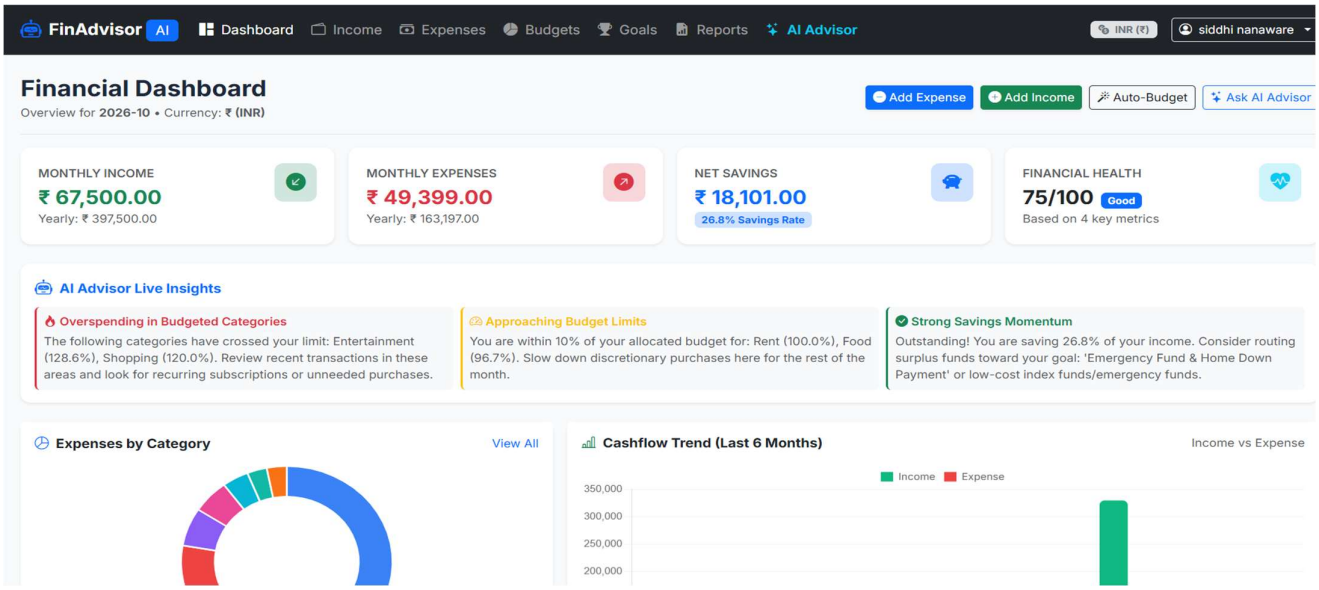
Data element	Attributes displayed in the interface
Income record	Date, source, description, amount
Expense record	Category (Rent, Food, Shopping, Education, Entertainment, Transport, Utilities, Healthcare, Bills); other attributes not shown
Category budget	Month, category, allocated limit, amount spent, remaining amount, utilization percentage, status
Monthly report data	Total inflow, total outflow, net savings, savings rate, financial health score, category and source breakdowns
Goal	A goal named “Emergency Fund & Home Down Payment” is referenced in the AI assessment; other attributes not shown

Income source types shown: **Salary, Business, Freelancing, Other**. Expense categories shown in the dashboard chart legend: **Rent, Food, Shopping, Education, Entertainment, Transport, Utilities, Healthcare, Bills**.

8.3 Schema Diagram

An entity-relationship diagram was not provided. Insert the diagram generated from the project's own models below.


FinAdvisor AI
Login
Register
You have been logged out successfully.
Welcome Back
Sign in to your Personal Finance Advisor
Email Address
name@example.com
Password
Enter password
Remember me on this device
Sign In
Don't have an account? Register here


FinAdvisor AI
Dashboard
Income
Expenses
Budgets
Goals
Reports
AI Advisor
INR (₹)
siddhi nanaware
Financial Dashboard
Overview for 2026-10 • Currency: ₹ (INR)
Add Expense
Add Income
Auto-Budget
Ask AI Advisor
MONTHLY INCOME
₹ 67,500.00
Yearly: ₹ 397,500.00
MONTHLY EXPENSES
₹ 49,399.00
Yearly: ₹ 163,197.00
NET SAVINGS
₹ 18,101.00
26.8% Savings Rate
FINANCIAL HEALTH
75/100 Good
Based on 4 key metrics
AI Advisor Live Insights
Overspending in Budgeted Categories
The following categories have crossed your limit: Entertainment (128.6%), Shopping (120.0%). Review recent transactions in these areas and look for recurring subscriptions or unneeded purchases.
Approaching Budget Limits
You are within 10% of your allocated budget for: Rent (100.0%), Food (96.7%). Slow down discretionary purchases here for the rest of the month.
Strong Savings Momentum
Outstanding! You are saving 26.8% of your income. Consider routing surplus funds toward your goal: 'Emergency Fund & Home Down Payment' or low-cost index funds/emergency funds.
Expenses by Category
View All
Cashflow Trend (Last 6 Months)
Income vs Expense
Income
Expense
350,000
300,000
250,000
200,000

9. Financial Calculations and AI Logic

9.1 Budget Utilization and Status

The Budgets page lets the user set a limit for each category and monitors spending against it. Its page description is “Set category limits, monitor real-time utilization, and prevent overspending.” For the month shown (September 2026), the page reports a total budget limit of ₹53,000.00, total spending of ₹110,800.00 (209.1% of the total budget) and a remaining budget of ₹-57,800.00, with 8 categories configured. The Auto-Generate (50/30/20) button creates a budget automatically; the allocation rule and category mapping behind it.

Each category carries a status label according to the threshold guide displayed on the page:

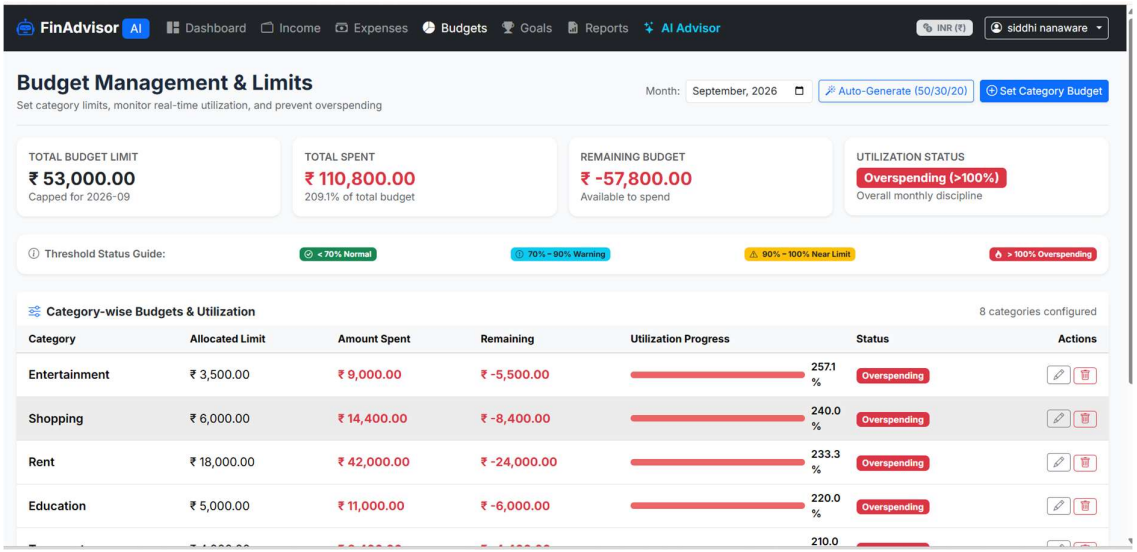
Utilization	Status label
Below 70%	Normal
70% to 90%	Warning
90% to 100%	Near Limit
Above 100%	Overspending

Utilization for a category is the amount spent divided by its allocated limit. For example, Entertainment shows ₹9,000.00 spent against a ₹3,500.00 limit, which equals the displayed 257.1%.

Table 9.1: Category budgets visible in Figure 9.1 (September 2026)

Category	Allocated limit	Amount spent	Remaining	Utilization	Status
Entertainment	₹3,500.00	₹9,000.00	₹-5,500.00	257.1%	Overspending
Shopping	₹6,000.00	₹14,400.00	₹-8,400.00	240.0%	Overspending
Rent	₹18,000.00	₹42,000.00	₹-24,000.00	233.3%	Overspending

Category	Allocated limit	Amount spent	Remaining	Utilization	Status
Education	₹5,000.00	₹11,000.00	₹-6,000.00	220.0%	Overspending



9.2 Monthly Financial Metrics

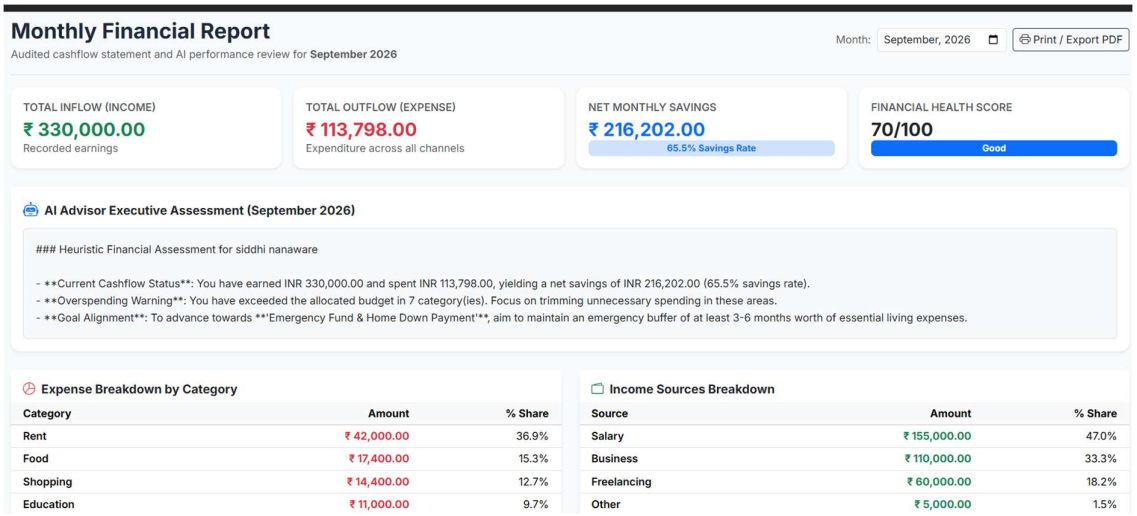
The Monthly Financial Report presents four headline figures for the selected month. The values shown for September 2026 are given below. Net savings equals total inflow minus total outflow, and the savings rate equals net savings divided by total inflow; the displayed figures are consistent with both relationships.

Metric	Value shown (September 2026)	Label in report
Total inflow (income)	₹330,000.00	Recorded earnings
Total outflow (expense)	₹113,798.00	Expenditure across all channels
Net monthly savings	₹216,202.00	65.5% savings rate
Financial health score	70/100	Good

The report also includes an expense breakdown by category and an income-source breakdown, each with amounts and percentage shares. The visible rows are shown below (the expense table continues beyond the captured area).

Expense category	Amount	Share	Income source	Amount	Share
Rent	₹42,000.00	36.9%	Salary	₹155,000.00	47.0%
Food	₹17,400.00	15.3%	Business	₹110,000.00	33.3%
Shopping	₹14,400.00	12.7%	Freelancing	₹60,000.00	18.2%
Education	₹11,000.00	9.7%	Other	₹5,000.00	1.5%

The Budgets page total spent (₹110,800.00) and the report total outflow (₹113,798.00) are reproduced as each page displays them. The financial health score formula and its rating bands.



9.3 AI Advisor Assessment

The report contains an “AI Advisor Executive Assessment” panel. For September 2026 it contains three items: a cashflow status statement (income earned, amount spent, net savings and savings rate), an overspending warning stating that the allocated budget was exceeded in seven categories, and a goal-alignment note for the goal “Emergency Fund & Home Down Payment” .

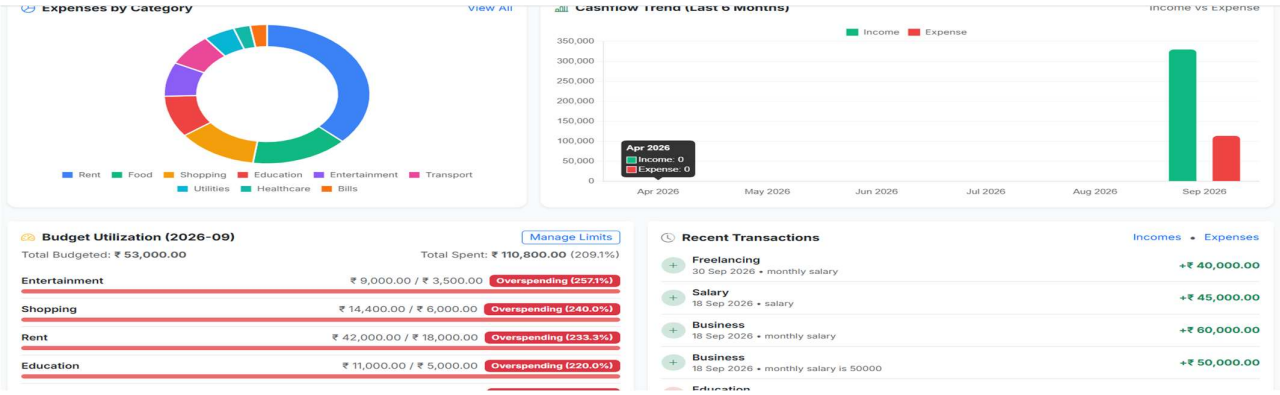
10.1 Interface Structure

The interface uses a dark top navigation bar displaying the FinAdvisor AI name and links to Dashboard, Income, Expenses, Budgets, Goals, Reports and AI Advisor. A currency indicator (INR, ₹) and a user account menu appear at the right of the bar. Pages use card-based summary tiles, tables with edit and delete actions, progress bars and colour-coded status labels. The frontend is written in JavaScript; frameworks and styling libraries.

10.2 Dashboard

The Dashboard combines four panels:

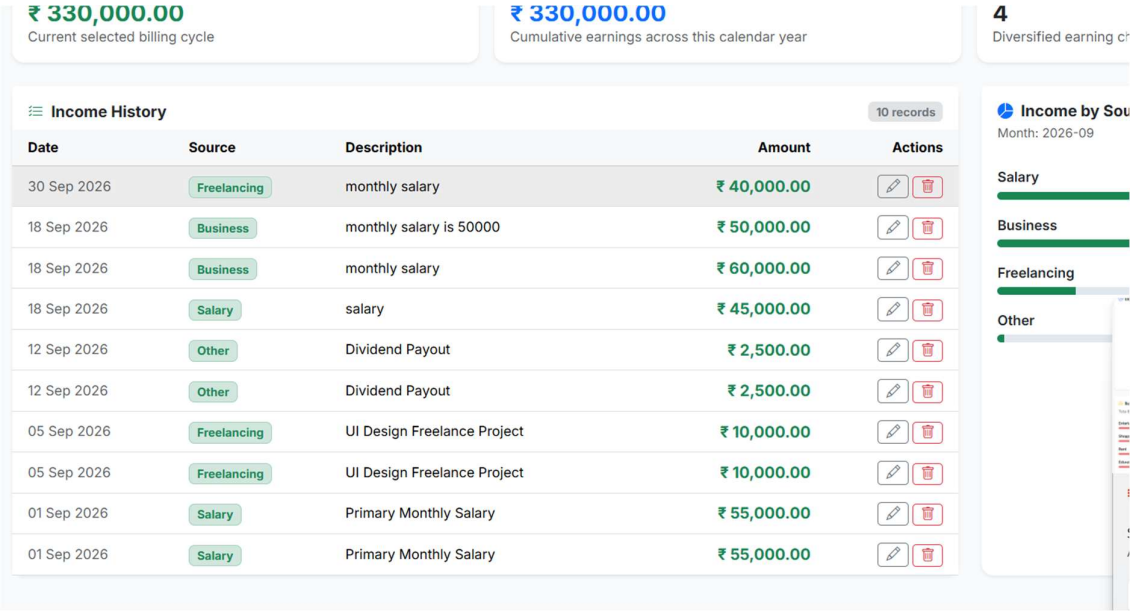
- **Expenses by Category:** a doughnut chart whose legend lists Rent, Food, Shopping, Education, Entertainment, Transport, Utilities, Healthcare and Bills, with a View All link.
- **Cashflow Trend (Last 6 Months):** a bar chart of income against expense from April to September 2026, with a tooltip showing monthly values.
- **Budget Utilization (2026-09):** total budgeted and total spent with per-category progress bars and overspending labels, plus a Manage Limits link.
- **Recent Transactions:** the latest income entries with links to the Incomes and Expenses lists.

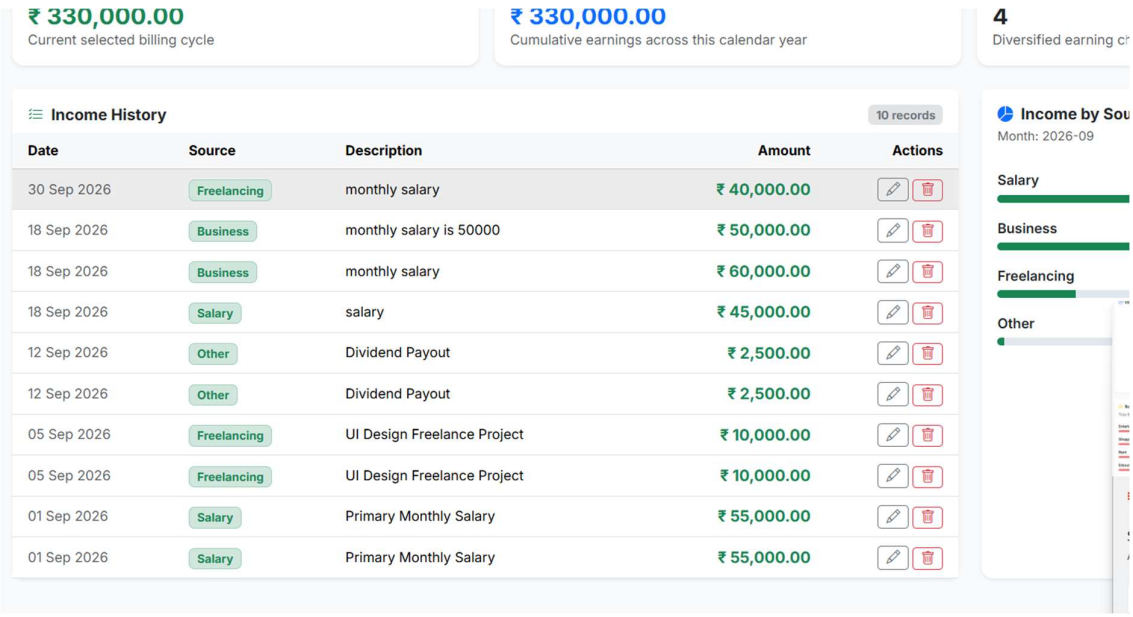


10.3 Income Page

The Income page opens with summary cards: ₹330,000.00 for the current selected billing cycle, ₹330,000.00 for cumulative earnings across the calendar year, and a third card showing the figure 4 under a label beginning “Diversified earning” (the

label is cut off in the screenshot). The Income History table lists each entry with its date, source, description and amount, together with edit and delete buttons, and reports 10 records. An Income by Source panel for 2026-09 shows progress bars for Salary, Business, Freelancing and Other.





11. Testing and Deployment

11.1 Testing

The testing approach, test cases and test results are The table below is a template to be completed from the project's own testing.

11.2 Deployment

The project requirements list the AWS CLI as a required tool and refer to AWS Labs and deployments in the hardware requirements. The AWS services used, the deployment procedure and the hosting configuration.

12. Project Milestones

12.1 Project Record

Item	Detail
Epics	8 (names: [Information not provided])
Tasks	15
Subtasks	0
Mentor	No mentor assigned yet

12.2 Module Milestones

Because epic and task names were not supplied, milestones are organised by functional module. Status reflects what the supplied material evidences. Replace this table with the actual epic names and dates if available.

No.	Milestone	Deliverable	Status	
M1	Income management	Income history, source categories, edit and delete, summary cards, income-by-source panel	Evidenced (Figure 10.2)	
M2	Expense management	Expense logging by category	Described in brief; page screenshot not provided	
M3	Budget management	Category limits, utilization status, Auto-Generate (50/30/20)	Evidenced (Figure 9.1)	
M4	Dashboard	Category chart, six-month cashflow trend, budget utilization, recent transactions	Evidenced (Figure 10.1)	

No.	Milestone	Deliverable	Status	
M5	Monthly Financial Report	Metrics, breakdowns, AI assessment, Print / Export PDF	Evidenced (Figure 9.2)	
M6	AI Advisor	AI assessment and recommendations	Assessment evidenced in report; dedicated page not provided	
M7	Goals	Goals module	Present in navigation; details not provided	
M8	Testing and deployment	Test results; deployment	pass	

12.3 Future Milestones

The proposed enhancements in Section 14 form the forward-looking milestones of the project. No schedule has been provided for them.

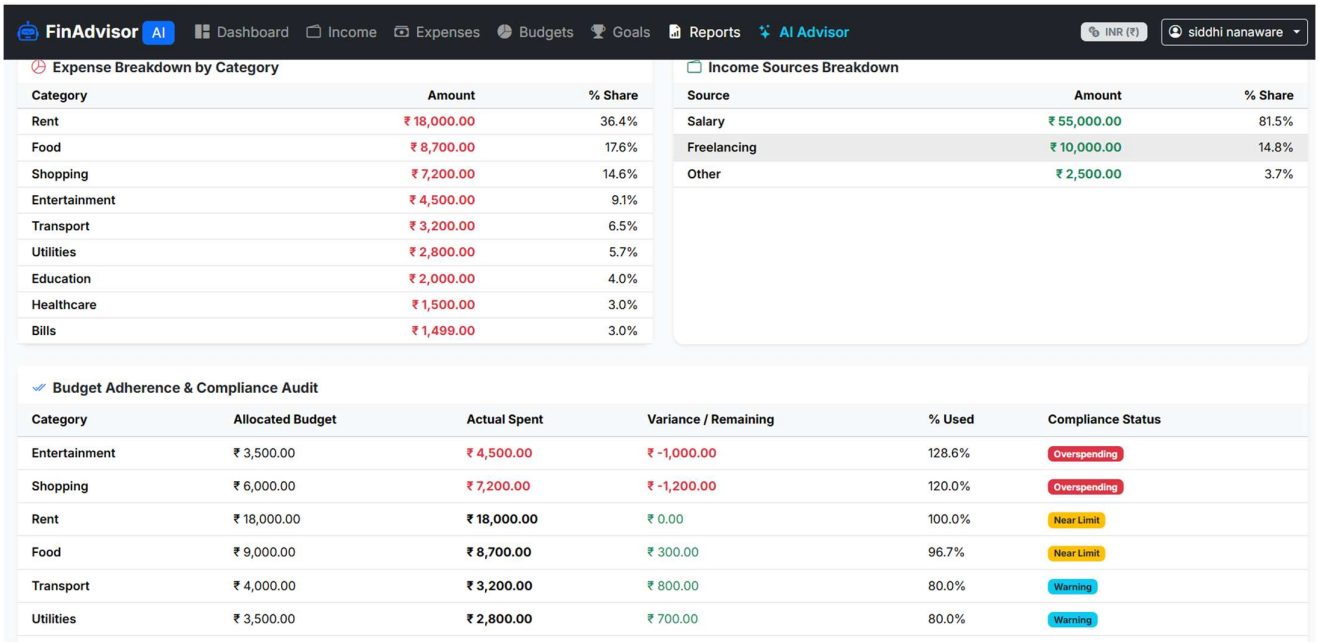
13. Screenshots, GitHub and Demo

13.1 Screenshot Index

Figure	Page	Section
Figure 6.1	System architecture diagram	6.1
Figure 6.2	Financial workflow diagram	6.3
Figure 9.1	Budget Management & Limits	9.1
Figure 9.2	Monthly Financial Report	9.2
Figure 10.1	Dashboard	10.2
Figure 10.2	Income page	10.3

13.2 Screenshots Still to Be Added

Replace each placeholder below with the corresponding screenshot. Delete any placeholder for a page that does not exist in the project.



13.3 GitHub Repository

Item	Detail
Repository link	https://github.com/SiddhiNanaware/personal_finance_advisor-

13.4 Demo

Item	Detail
Demo video / live link	https://drive.google.com/file/d/1zwwCtoxWss-kmmihLc7XGFx_b9PYSESP/view?usp=sharing

14. Conclusion and Future Enhancements

14.1 Conclusion

The Personal Finance Advisor Bot (FinAdvisor AI) gives individuals one environment for recording income and expenses, setting and monitoring category budgets, and reviewing a structured monthly report with an AI-generated assessment. Built on Flask, SQLAlchemy, SQLite, JavaScript and GeminiAI, it replaces manual tracking with automated utilization status, net savings and savings-rate figures. The four scenarios in the project brief show that the same design serves salaried professionals, students, freelancers and households, and the layered architecture leaves room for the enhancements below.

14.2 Future Enhancements

The following features are proposed and are not part of the current functionality.

Proposed enhancement	Description
Predictive spending analytics	Forecasting of future spending patterns
Investment suggestions	Recommendations related to investing
Goal-based savings tracking	Tracking of savings progress against defined goals
Advanced AI-driven financial health monitoring	Deeper AI-based assessment of overall financial health

15. Student Contribution and Learning Outcomes

Project focus

This project focuses on making everyday money management easier through one place to record income, track expenses, set category budgets and review monthly financial results.

Key learning outcomes

The project brings together frontend interaction, a Python Flask application, SQLAlchemy database access and SQLite storage. It also demonstrates how an AI Advisor can present a readable assessment based on financial information.

Practical value

Budget utilization and status labels help users notice when spending approaches or exceeds a limit. Monthly metrics such as total income, total expenses, net savings and savings rate provide a compact view of financial activity.

Responsible use and future improvement

Financial records should be handled carefully and kept private. AI-generated suggestions should be treated as general guidance rather than a substitute for advice from a qualified financial professional. Future work could add stronger input validation, privacy controls, goal tracking and testing with sample data.